1 Computing System

"Computer is not a device any more. It is an extension of your mind and your gateway to other people."

Mark Shuttleworth.

1.1 Introduction

The journey of computing devices spans thousands of years from simple counting tools to modern supercomputers and smart phones. Over a relatively short period, computing devices have progressed from performing basic calculations to handling complex data processing tasks with incredible speed. Computers today are fast, compact and powerful. But they were not always like this. This chapter covers the historical evolution, technical organisation, concepts of data processing, software classifications, importance of open source software, green computing and issues of e-waste. Studying this evolution and other aspects helps us to realise the development of technology and its ongoing influence on our lives. Now the development is like '*We dream in the morning, a reality by noon and obsolete by evening*'.

1.1.1 Evolution of Computing

The word compute means to 'calculate'. Computing devices have transformed from basic calculators to advanced data processors in just a few decades. The *Abacus* is considered to be the first known calculating tool. It uses beads on rods to perform basic arithmetic operations like addition and subtraction. It has been in use for thousands of years across different cultures. *Charles Babbage*, who is known as the '*Father of the Computer*', designed a mechanical general-

purpose computer in 1837. Although it was never fully built in his lifetime, the design laid the foundation for modern computing. In 1843, *Augusta Ada Lovelace*, wrote the first algorithm intended for a machine (Charles Babbage's Analytical Engine). She is therefore often regarded as *the first computer programmer*. Although the Analytical Engine was never built in her lifetime, her contributions laid the foundation for the development of computational theory. In 1890, *Herman Hollerith* invented a punch-card-based machine used in the U.S. census. It marked the shift towards data processing. The evolution of computing devices is shown in Figure 1.1

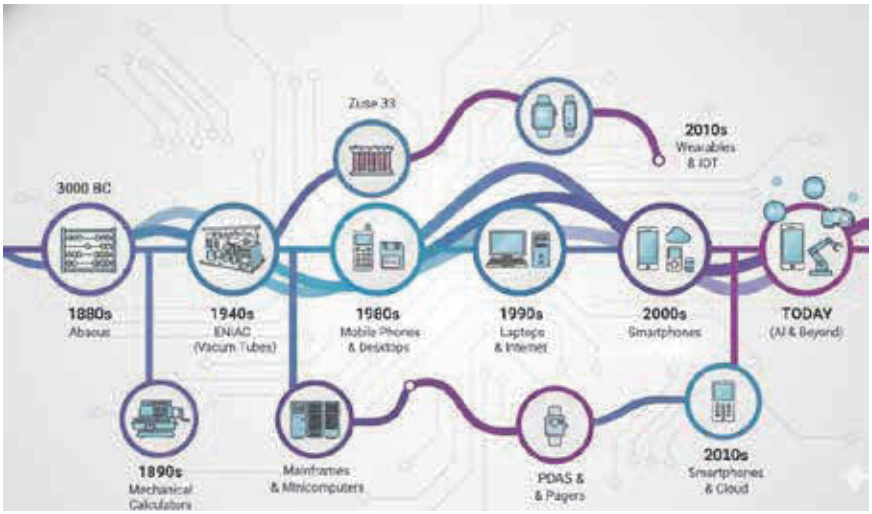


Figure 1.1 - Timeline diagram of generation of computing devices

1.1.2 Generations of Computing

The *Generations of Computing* refers to the classification of computers based on major technological developments over different periods. These developments increased the speed, reduced the size and cost, and enhanced the efficiency of computers. The evolution of computers is classified into *five generations*.

i) First Generation (1940–1956)

This era marks the beginning of modern electronic computing. The technology used was *vacuum tubes* for circuitry, and magnetic drums for memory. These tubes were easily broken, generated a lot of heat, and consumed a huge amount of power. Computers were bigger in size, often occupied the entire room. Programming was done with the help of machine language in which instructions are written using 0s and 1s. *ENIAC* (Electronic Numerical Integrator and Computer) is considered to be one of the first general-purpose electronic computers introduced in 1945. It was used by the U.S. Army for calculating ballistics tables. *UNIVAC I* (Universal

Automatic Computer)was the first commercial computer produced in the United States. It was used to predict the outcome of the 1952 presidential election on live television. Another milestone in this period was *EDSAC* (Electronic Delay Storage Automatic Calculator), one of the earliest stored-program computers, designed and built at the University of Cambridge in England. It became operational in 1949 and played a crucial role in the development of computing.

ii) Second Generation (1956–1963)

The invention of the transistor replaced vacuum tubes in computers. Transistors were much smaller, more reliable, and more energy-efficient than vacuum tubes. This generation saw the rise of the computer industry and the start of widespread use of commercial computers. High level programming languages like COBOL and FORTRAN were developed, making programming easier and more accessible. Computers became smaller in size, and new machines *IBM 1401* and *IBM 7090* were also introduced in this period.

iii) Third Generation (1964–1971)

This generation is defined by the *integrated circuit* (IC), which was invented by Jack Kilby and Robert Noyce. It is a small chip, also known as microchip, containing thousands of transistors and other components. Computers became even smaller, faster and more affordable, leading to their use in a wider range of industries. Operating systems were introduced, allowing computers to run multiple programs simultaneously and manage resources more efficiently. IBM 360 series machines were the newly introduced computing device in this era.

iv) Fourth Generation (1971–Present)

The development of the *microprocessor*, a single IC that contains all the functions of a computer's central processing unit (CPU), marked a turning point. This generation brought computing to the general public, transforming society and giving rise to the digital age. The microprocessor became the brain of the computer. The invention of the *Intel 4004* microprocessor in 1971 is a landmark event. The development of more user-friendly software and the rise of the Graphical User Interface (GUI) made computers accessible to the general public. The invention of the CPU led to the creation of personal computers (PCs), laptops and eventually mobile devices. The internet was also developed during this era. The important machines introduced were *Apple II, and IBM PC*.

v) Fifth Generation (Present and Beyond)

While the fourth generation is still prominent, the concept of fifth generation is often used to describe future and present technologies that go beyond traditional computing. We are currently witnessing and utilising the parallel or mixed use of fourth and fifth-generation computers. This generation aims to create computers that can interact with the world in more natural ways and solve problems of incredible complexity, shaping the future of science, medicine and technology. Technologies like Artificial Intelligence (AI), Parallel Processing, Quantum Computing and DNA Computing have been incorporated in to the computing process of this era. Currently, IoT (Internet of Things), Modern AI systems and quantum computers are also in development stage.

- **Artificial Intelligence (AI):** It is the ability of a computer or machine to perform tasks that normally require human intelligence, such as learning, reasoning, linguistic intelligence, perception and problem-solving.
- **Parallel Processing:** It is the use of multiple processors working together to solve a single problem, increasing speed and efficiency.
- **Internet of Things (IoT):** The Internet of Things (IoT) refers to a vast network of physical objects (things) that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.
- **Quantum Computing:** A new approach that uses the principles of quantum mechanics to perform calculations, with the potential to solve problems that are currently impossible for traditional computers.
- **DNA Computing :** This is a form of biological computation that uses DNA molecules, enzymes, and biological reactions instead of traditional silicon-based transistors to process information.



1. The first known calculating device is _____.
2. The second generation of computers used _____ instead of vacuum tubes.
3. _____ is known as the father of computers.
4. The second generation of computing introduced _____ programming language.

Try Yourself

5. GUI was initially used in _____ generation of computers.
6. What are the contributions of Ada Lovelace in computing.
7. Distinguish between first and second generation computers.
8. Expand ENIAC, EDSAC and UNIVAC.
9. What is artificial intelligence?
10. Identify the main technology of each generation of computing

Match the following

1. First Generation	a) Artificial Intelligence
2. Second Generation	b) Vacuum Tubes
3. Third Generation	c) Microprocessors
4. Fourth Generation	d) Transistors
5. Fifth Generation	e) Integrated Circuits

1.2 Computer System

Computer is an electronic data processing machine that can be programmed to input, store and process data to produce output. It is made up of different hardware and software components.



Figure 1.2 - Computing devices

Computer system usually consists of hardware components such as input/output (I/O) units, Central Processing Unit (CPU) and Storage Unit. All these components are mutually dependent and work together as a single unit to produce the desired output. It becomes a system when the different units perform their individual tasks to achieve a common goal. Nowadays, computers come in various forms such as servers, personal computers (PCs), laptops, tablets, and smart phones. Some of the devices are shown in Figure 1.2.

1.2.1 Computer Organisation

A computer is an electronic device that processes raw data into meaningful information. This transformation is made possible through the organised arrangement of various hardware components, known as functional units. A computer operates using both hardware and software, working together to perform the tasks efficiently. Its internal organisation is designed to ensure the smooth flow of data from input devices to processing units, storage components, and finally to output devices. Computer organisation refers to the structure, interconnection, and operation of these hardware components that enable the execution of instructions. This session focuses on understanding the hardware components of a computer and how they are organised to function as an integrated system.

The computer system is organised into different functional units that perform specific tasks. These units work together to achieve the overall purpose of data processing. The main *functional units* are *Input Unit, Central Processing Unit (CPU), Storage Unit (Memory) and Output Unit*. The basic structure of a computer system can be broken down into three main categories - Hardware, Software and User. Hardware refers to the physical, tangible components of the computer. Software is the set of instructions and programs that tell the hardware to perform a task. User is the one (Human, Device or AI tool) who interacts with the computer system.

1.2.2 Block Diagram of Computer Organisation

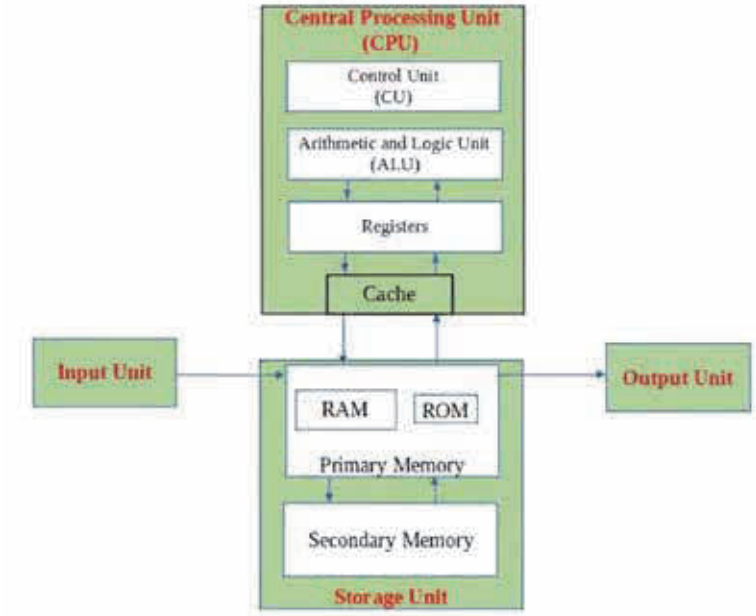


Figure 1.3 - Block Diagram of Computer Organisation

The Von Neumann architecture, published by John Von Neumann in 1945, is the foundational design for almost all modern computers. It is based on the stored-program concept, meaning that both instructions and data are stored in the same memory unit. The block diagram is depicted in Figure 1.3.

i) Input Unit

The input unit is used to feed data into a computer system. Input devices accept data and instructions, and convert them into digital form that a computer can understand. The data is encoded into machine-readable form (binary) and is transferred to the processing unit. Some of the input devices are keyboard, mouse, scanner, microphone, barcode reader, touch screen, etc. Specially designed input devices are commonly available today. Besides, we can input data through voice, visuals, biometric sensing, etc. Voice search and mobile biometric passwords are some examples.

Computing has evolved far beyond the traditional usage of keyboard and mouse. Nowadays, we have shifted to '*multimodal*' control. This simply means you can pick the easiest way to talk to your devices using your *voice, touch, eyes, or even your thoughts*. Modern gadgets are now '*ambient*'. That means, '*ambient gadgets*' are devices that work quietly in the background of your life you don't have to think about them or control them like a traditional computer. Instead of a computer sitting on a desk, technology is built into our clothes, our bodies, or the rooms around us. For example smart lights, smart watches and voice assistants. Because these devices are getting smarter, we have changed how we use them to make life faster and easier for everyone. The biggest trend right now is the '*Always-Available*' strategy. Think of it like the air around you, it is always there, even if you cannot see it. We use small sensors worn on our body so that we can control your digital world instantly from anywhere. Figure 1.4 depicts some input devices and modern strategies.

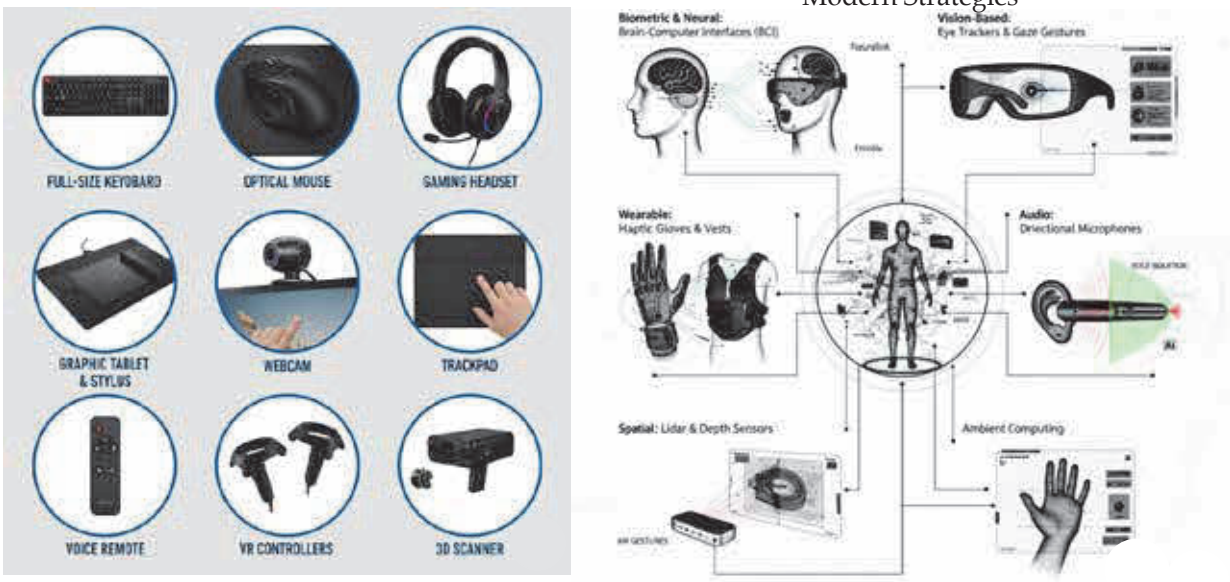
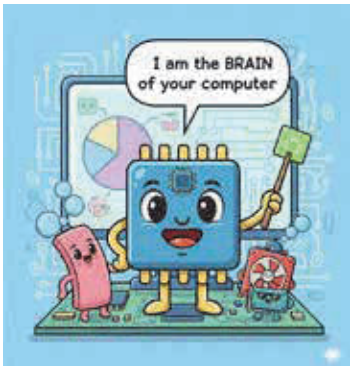


Figure 1.4 - Input devices and modern strategies

ii) Central Processing Unit (CPU)

It performs all the processing, and controls the functions of other units inside the computing device. CPU is commonly known as the processor. It consists of several microchips usually called Integrated Circuits (IC).

The CPU works based on the instructions given through programs. It fetches data and programs from memory and performs arithmetic and logical operations as per the given instructions to produce results, which are then stored back into memory. During processing, the CPU stores data and instructions in high-speed memory called registers, which are a part of the CPU.



CPU

Apart from registers, the CPU has two more main components -Arithmetic Logic Unit (ALU) and Control Unit (CU). CU acts like a '*traffic police man*' who controls the traffic system, inside the computer. It directs the flow of data between different functional units and interprets instructions in the programs. ALU performs all arithmetic operations like addition, subtraction, multiplication, division and executes logical operations like comparisons such as greater than, equal to, or less than, etc. This component has the main role in decision-making processes.

Registers are the smallest and the fastest storage locations within the CPU. They temporarily hold data, instructions, and memory address necessary for the CPU to perform operations. CPU registers are categorized based on their specific function in the instruction execution cycle.

The two main categories are *Special-Purpose Registers (SPRs) and General-Purpose Registers (GPRs). Special-Purpose Registers (SPRs)* are essential for managing the state and control of the CPU and the program's execution flow. General-Purpose Registers (GPRs) are flexible and can be used by the programmer (via assembly language) or the compiler to store temporary data and intermediate results during the execution of a program. The different SPRs and GPRs are listed in Table 1.1. We will discuss more about microprocessors in section 1.2.3.

Table 1.1 - SPRs and GPRs

Special-Purpose Registers (SPR)	General-Purpose Registers (GPR)
Program Counter (PC)	AX (Accumulator)
Instruction Register (IR)	BX (Base)
Memory Address Register (MAR)	CX (Counter)
Memory Buffer Register – MBR(also called Memory Data Register-MDR)	DX (Data)
Status Register	SI (Source Index)
Stack Pointer (SP)	DI (Destination Index)

iii) Output Unit

The *output unit* is responsible for converting the processed data from the computer's digital format into human-readable format. It is used to present the results after processing the data. The monitor, projector, headphone, speaker and printer are some examples. Visual display Unit (VDU) like monitor and projector are the most commonly used *softcopy* output devices, and printer is the most widely used hardcopy output device. Inkjet, laser jet and dot matrix are the widely used printers for office purpose. Recently, 3D printers are also available for 3D design printing to build exact physical replicas of 3D images. 3D printing has a wide range of uses in prototyping and manufacturing industries like automotive, aerospace, medical, architectural and construction. Some of the output devices are shown in Figure 1.5.



Figure 1.5 - Output devices

iv) Storage unit

The computer system needs to store data and instructions before processing and the results after processing. The storage unit or memory is where data and instructions are held for future use. It is essential to store data temporarily or permanently. At the physical

level data is stored in the form of 0s and 1s. The binary digits 0 and 1 are the basic units of memory called *bits* (**bi**nary digi**ts**). A 4 bit combination of bits is known as *nibble* and eight bit combination is called *byte*. In computing, memory size is measured using a hierarchy of units based on the number of bits. Table 1.2 contains the different memory units and their size.

Table 1. 2 - Memory units and their size

Unit	Abbreviation	Size
Bit	b	1 or 0 (1 bit)
Nibble	N (or nb)	half byte (4 bits)
Byte	B	1 Byte (8 Bits)
Kilobyte	KB	1,024 bytes
Megabyte	MB	1,024 KB
Gigabyte	GB	1,024 MB
Terabyte	TB	1,024 GB
Petabyte	PB	1,024 TB
Exabyte	EB	1,024 PB
Zettabyte	ZB	1,024 EB
Yottabyte	YB	1,024 ZB

Storage unit is generally classified into Primary Storage (Main Memory) and Secondary Storage (Auxiliary Memory). Primary Memory is also called internal memory which is directly accessible by the CPU, and used for short-term storage. RAM (Random Access Memory) is the most common type of primary storage. It is fast but volatile, in the sense that data is lost when the power is turned off. ROM (Read Only memory) and Cache memories are also considered as primary memory due to their direct accessibility by the CPU. RAM and cache are volatile, whereas ROM is non-volatile. RAM is used to hold the data and programs that are being used by the computer system while running. ROM stores essential programs such as the BIOS (Basic Input Output System) and other firmware, which are necessary to start a computer. For example, when we press the power button of a computer, the bootloader stored in to ROM initiates and loads the operating system from secondary storage into RAM. This process is called *booting*.

Boot Loader

A boot loader is a small program that runs every time you power on a computer, smart phone, or any other computing device. Its primary role is to load the operating system into RAM, a process that follows the Power-On Self-Test (POST).

The *Cache* memory is a small, high-speed memory located inside or very close to the CPU. Usually it is placed in between RAM and CPU. Even though RAM is faster than secondary memory, it is not as fast as CPU. Due to this slowness of RAM, CPU will be idle for an interval of time without getting data from RAM. This may lead to the lack of maximum utilisation of CPU. To avoid this delay, Cache memory is used with CPU. It stores copies of frequently accessed data from RAM. When the CPU needs data, it first searches the cache memory. If the data is in the cache, it is read from there; otherwise, it is fetched from RAM. Different types of Cache memories in the CPU are L1 (Level 1), L2 (Level 2) and L3 (Level 3).



Memory hierarchy

Secondary Storage or External Memory is used for long-term storage of data and programs. It is non-volatile, slower and has a larger capacity than primary storage. Some secondary storage devices are Hard Disk Drives (HDDs), Solid-State Drives (SSDs), USB flash drives, and memory cards. Nowadays, comparing with HDD, SSD is more common in use because it is much faster, more durable, has less noise, lower power consumption, lightweight and compact. The emerging technology like *quantum computing* may change the existing memory devices. Different storage devices are shown in Figure 1.6.



Figure 1.6 - Storage devices

1.2.3 Microprocessor

A microprocessor is a computer processor where the data processing logic and control are included on a single integrated circuit (IC), or a microchip. Figure 1.7 shows the image of a microprocessor. Moore's Law, as stated by Intel co-founder Gordon Moore in 1965 (revised in 1975), is the observation that the number

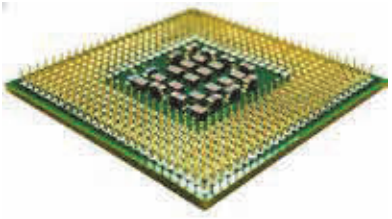


Figure 1.7 - Microprocessor

of transistors that can be economically placed on an integrated circuit (IC) doubles approximately every two years. In the early 1960s, SSI (Small Scale Integration) chips were manufactured with approximately 1 to 100 transistors. By the early 1970s, technology moved to MSI (Medium Scale Integration), where between 100 and 1,000 transistors were integrated onto a single chip. Later in 1970s,LSI (Large Scale Integration) chips that consist of 1,000 to 10,000 transistors, in 1980s,VLSI (Very Large Scale Integration) chips that use 10,000 to millions of transistors, and at present ULSI (Ultra Large Scale Integration) chip comprising millions to billions of transistors are introduced. These developments are shown in Figure 1.8.

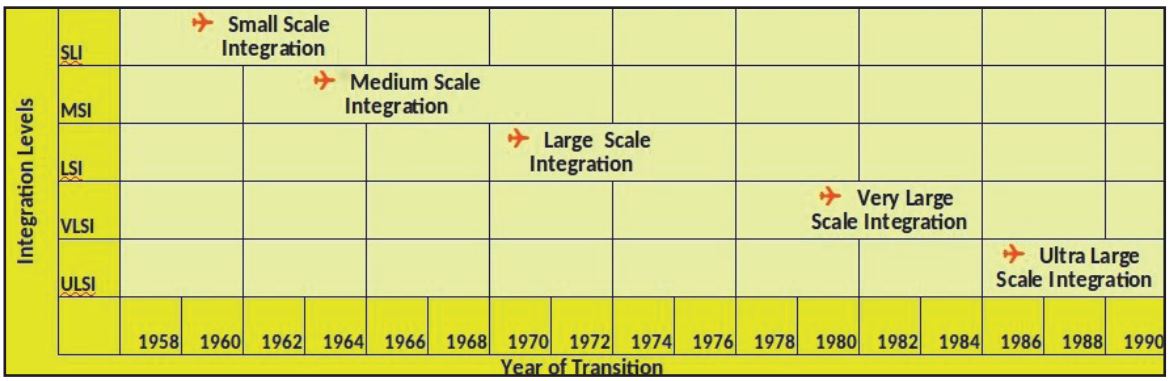


Figure 1.8 - Timeline diagram of micro processor evolution

1.2.4 Performance Characteristics of Microprocessors

The microprocessors are used based on several key criteria that reflect their architecture, performance, and intended applications. The followings are some performance characteristics of microprocessors.

i) Word Length

This refers to the number of bits the microprocessor can process at a time. Table 1.3 shows microprocessors with various word lengths.

Table 1.3 Microprocessors with different word lengths

Microprocessor	Word Length
Intel 4004	4-bit
Intel 8085, Zilog Z80	8-bit
Intel 8086, Motorola 68000	16-bit
Intel 80386, ARM7	32-bit
Intel Core i7, AMD Ryzen	64-bit

ii) Clock Speed

Clock speed simply means the number of pulses generated per second by the clock inside a computer. The clock speed indicates the speed at which the computer can execute instructions. Earlier, it was measured in Hertz (Hz) and Kilohertz (kHz). But with advancement in technology and chip density, it is now measured in Gigahertz (GHz), i.e., billions of pulses per second.

iii) Number of Cores

Imagine an office with only a single worker responsible for every task. If there are two or more workers, tasks can be performed in parallel, resulting in faster job completion. Think of a single-core CPU as an office with just one worker handling every job sequentially. Adding more workers (cores) allows tasks to be done in parallel, leading to much faster performance. Thus we can say core is the physical processing unit within a microprocessor (CPU) that executes instructions. Today we are using multi-core processors such as Dual-core, Quad-core, Octa-core, Hexa-core etc., in modern CPUs.

1.2.5 Motherboard

A motherboard is the main circuit board of a computer. The motherboard acts as the backbone of the computer system, providing power, connections, and communication paths for all hardware components to work together. It is the main printed circuit board (PCB) that connects and allows communication between all major components, such as

- Processor
- RAM ,ROM
- Storage devices like HDD,SSD, etc.
- Input/Output devices (keyboard, mouse, etc.)
- Expansion cards (graphics, sound, network cards)



Motherboard



1. _____ is called the brain of a computer.
2. _____ memory stores data permanently.
3. The main function of _____ is to perform arithmetic and logical operations.
4. ULSI stands for _____ .
5. 1KB = _____ bytes.
6. Explain the role of Control Unit.
7. What is the importance of cache memory?



8. What is word length?
9. List any six storage devices.
10. Name any six input and output devices.
11. Compare RAM and ROM.
12. Differentiate between Primary and Secondary memories.
13. Differentiate between softcopy and hardcopy outputs.
14. What are registers? List the registers inside CPU.
15. Explain core in a microprocessor.
16. Expand SSI, MSI, LSI.
17. Explain the working of a computer system with the help of a block diagram.
18. Explain the role of motherboard. List any two components connected with it.

Match the following

1. Input Unit	(a) Monitor
2. Output Unit	(b) Keyboard
3. ALU	(c) RAM
4. Storage	(d) Arithmetic operations

Activity

1. Collect the latest computing device configurations from advertisements in periodicals.(Paper cuttings or written documents)
2. Identify different input, output and storage devices in computer lab.

1.3 Software

Computer system is a combination of hardware and software. We are already familiar with the physical parts or hardware of computer system. Hardware of a computer system is nothing unless it is operated by instructions. Software is a set of instructions (programs) that tell the computer to perform an action. Without software, a computer is just an inert machine. Some examples of software are operating systems like Linux, Windows and Android, word processing tools like LibreOffice Writer and Microsoft Word,

Spreadsheets like LibreOffice Calc, Presentations like impress and Powerpoint, video players like VLC Player, photo editors like GIMP and database applications like Base, MS Access, MySql and MariaDB. Based on the mode of interaction with hardware and functionality, software is broadly classified into two types: System Software and Application Software.

1.3.1 System Software

System software is a set of programs designed to control and manage the computer hardware. System software manages the overall functioning of the computer system. It acts as a bridge between the hardware and the user's applications. Without system software, application software cannot run. Examples of system software include operating systems, system utilities, device drivers, firmware, language translators, etc.

i) Operating Systems (OS)

An *Operating System (OS)* is an interface between user and the hardware. It is the main control program that manages all other programs in a computer. The OS is the first software loaded into the computer's memory when the system boots up. Table 1.4 shows the functions of Operating System.

Table 1.4 - Functions of an OS

Functions	Description
Memory Management	Allocating and de-allocating memory for running programs
Process Management	Managing the execution of all tasks and processes
File Management	Organizing and storing files on storage devices
Device management	Controlling all I/O devices such as keyboard, mouse, printer
Security and Access Control	Protecting data and system resources from unauthorized access
User Interface	Providing a way for the user to interact with the computer

Some common examples of Operating Systems are

Desktop/Laptop : Microsoft Windows , MacOS, Linux.

Mobile Phones : Android, iOS.

ii) Utilities

Utility software (Utility program) is used to manage, maintain, and optimize the computer system's performance, security, and disk organization. They are the tools that provide support for the smooth functioning of a computer. Some examples are antivirus software, disk clean up tools, file compression software, backup tools, disk defragmenter, etc. The different utility programs and their uses are shown in Table 1.5.

Table 1.5 - Utility programs and their use

Utility Software	Purpose	Examples
Antivirus/Anti-Malware	Protects the computer system from viruses, spyware, ransomware, and other malicious software	Windows Defender, Avast, Avira, McAfee, Malwarebytes, ClamAV/ClamWin, Panda
File Compression/ Decompression Tools	Reduces the size of files to save storage space and make files easier to transfer (e.g., via email)	WinRAR, 7-Zip, WinZip, PeaZip, gzip, bzip2,LZ4
Disk Management Tools	Manages the computer's hard drive, including partitioning, formatting, and analyzing disk space	Windows Disk Management, MiniTool Partition Wizard, GParted
Disk Cleanup Tools	Scans the hard drive for unnecessary files (like temporary files, cache, and log files) and removes them to free up disk space	Windows Disk Cleanup, CCleaner
Disk Defragmenters	Reorganizes fragmented data on a hard disk drive to store pieces of each file close together, improving data access speed	Windows Disk Defragmenter (or Optimize Drives), MyDefrag
Backup and Recovery	Creates copies of important data (files, folders, or entire systems) and allows for restoration in case of data loss or system failure	Windows Backup and Restore, Time Machine (macOS), Acronis True Image

Uninstallers	Completely removes unwanted software and all related files and registry entries from the system	Revo Uninstaller, IObit Uninstaller
System Monitoring/ Profiler	Provides real-time information and detailed reports on the system's performance, resource usage (CPU, RAM, disk), and hardware/software inventory	Windows Task Manager, Activity Monitor (macOS), CPU-Z

iii) Device Drivers

Device driver or Driver software is a software component that allows the operating system to communicate with the hardware devices in a computer. It works as a translator and a bridge between the operating system and hardware devices by converting commands into instructions that the device can understand and send data back to the operating system. Manufacturers provide these drivers and keeping them updated is crucial for performance, stability, and accessing all device features. The device driver specifications are listed in Table 1.6.

Table 1.6 - Examples of device drivers

Device Type	Driver Function
Printer	Tells the OS how to send print jobs
Graphics Card	Enables screen display, resolution, 3D support
Sound Card	Handles audio input/output
Wi-Fi Adapter	Manages wireless network connections
Touchpad	Provides multi-touch gesture support
USB Ports	Manages connected devices like pen drives

1.3.2 Application Software

Application software, often called an '*app*' is a program designed to perform a specific task or solve a particular problem for the user. It operates between the user and the system software. The figure shows the hierarchy of software and hardware. A computer system cannot work without *system software*, but it can function without *application software*. For example, without an operating system, a computer or mobile phone cannot run. However, even without



Hierarchy of software and hardware

application software such as a calculator, graphics editor, word processor, or spreadsheet, a computer or smart phone can still function. Some application software are web browsers, word processors, computer games, media players, and presentation tools. Application software is classified as follows.

i) General-purpose Software

The application software developed for generic purposes is called General-purpose software. They are usually used for performing common tasks. So many types of software like Word Processors, electronic spreadsheets, Presentations, Database Management System (DBMS), Web Browsers, Graphics Software are available in this category.

ii) Special-purpose Software

The application software exclusively designed to perform a specific task is called Special-purpose software. Some of the examples are Accounting Software, Reservation Systems, Inventory Control Systems, Payroll Systems and Media Players.

1.3.3 Programming Tools

A *program* is a set of instructions that a computer can execute to perform a specific task. *Programming* is the process of writing, testing, and maintaining these instructions. Programs are written in a specific *programming language*. Programming tools, also known as software development tools, are used to create, debug, test, and execute programs. They are essential for the entire software development process. As we know, if the computer wants to do a particular task, we need to give some instructions to process the input data for the desired output. We can communicate with each other using natural languages like Kannada, Malayalam, Tamil, etc. Humans can understand different languages but computers can understand only machine language (Binary language of 1s and 0s). Programmers write instructions in high level language and computers store and process them in machine language. So there must be a translator for the conversion. They are usually called language translators. To write instructions, code editor like Geany, IDLE or Eclipse is required. Here, we will briefly describe some of the program development components such as Programming languages, Language translators and Program development tools.

i) Programming Languages and Language Translators

Writing instructions in the form of 1s and 0s is a tedious task for the human being. To simplify the difficulties while writing instructions, different types of computer programming languages are introduced. These languages are basically classified into two types, Low-level and High-level languages. Low-level languages include machine language and assembly language, and they are machine dependent. 1s and 0s are used in machine language to write instructions which are directly understood and executed by the computer system. As we mentioned earlier, writing code in machine language by remembering operation codes, character codes and machine addresses is very difficult. Error detection and correction in the code is also difficult. Since the code written in machine language is directly understandable by the computer system, there is no need of a translator.

Assembly language was developed to simplify the code writing by using English like notations as symbolic opcodes and mnemonics instead of 1s and 0s. The main problem of writing code in this language is that, it is totally machine specific or machine dependent. The code written for one machine cannot be executed by another machine. Here the code written in assembly language cannot be directly understood by the computer and hence there should be a translator. The language translator called Assembler will translate a program written in assembly language into machine language.

Opcode(Operation Code)

An opcode is the portion of machine language instructions that specifies the operations to be performed. It is the only language represented in binary, a CPUs control unit understands.

Mnemonics

A mnemonic is an abbreviated name used to represent an opcode in assembly language. Instead of memorizing strings of binary digits, programmers use short words like ADD, MOV and JMP.

Writing and executing codes in low-level language is very difficult because they are totally machine dependent. To reduce and simplify the coding complexity, high-level languages were introduced. They are machine independent and use English-like sentences. The code written in High-level language is not directly understood by the computer system. Hence, translators are needed to convert high-level language code into machine language code.

The language translators, *compiler* and interpreter are used for this case. A compiler translates the entire high-level language program into machine code all at once before the program is executed (GNU Compiler Collection (GCC), Java compiler, etc.). An *interpreter* translates and executes the high-level language program line by line during runtime (Python interpreter, BASIC interpreter, etc).

ii) Program Development Tools

To write the program codes we need some text editors. An editor is a software that allows us to create a text file when we type instructions and store the file as the source code. Lightweight text editors are specifically designed for coding, offering features like syntax highlighting and code completion. In order to simplify the programming procedure we usually use a single screen environment. These are commonly known as Integrated Development Environments (IDEs). This environment integrates all necessary tools (editor, compiler, debugger, etc.) into a single, comprehensive interface. Python IDLE, Geany, Netbeans, Eclipse, Atom and Lazarus are examples for IDEs.

1.3.4 Free and Open-Source Software (FOSS)

Software that grants users the freedom to run, study, change, and share the software and its source code is called *Free and Open-Source Software (FOSS)*. We use FOSS everyday through different computing devices. Operating Systems such as Linux (e.g. Ubuntu, Fedora, Android are based on the Linux kernel), Web Browsers like Mozilla Firefox, Office Suites such as *LibreOffice (an alternative to* Microsoft Office*), Graphics & Design software like GIMP (GNU Image Manipulation Program, an alternative to Photoshop), Blender (3D creation suite), Web Servers like Apache HTTP Server and Media Players such as VLC Media Player are some examples of FOSS.*

The Free Software Foundation (FSF) is a global movement of individuals who support free software that promotes values and ethics. FSF was founded by Richard Stallman in 1985. *Richard Stallman* is co-founder of the GNU project and the free license movement, which has given rise to the biggest digital commons: GNU/Linux, Mozilla Firefox, LibreOffice, Wikipedia and Creative Commons. The word 'Free' refers to freedom, not price free. You can join the FSF community by becoming a Free Software Supporter. KITE (Kerala Infrastructure and Technology for Education) is a promoter of FSF activities.



Richard Stallman

i) Free Software

Free software refers to software that grants users the freedom to run, study, modify, and distribute it. It is important to note that *open-source* and *free software* are usually the same things, just with different goals. The goal of the free software movement is *social justice* and *user liberty*, whereas the open source movement focuses on *better software development*. The following are the four essential freedoms for free software.

1. **Freedom 0** :-Freedom to run the program for any purpose.
2. **Freedom 1** :-Freedom to study how it works and changes it.
3. **Freedom 2** :-Freedom to redistribute copies.
4. **Freedom 3** : Freedom to distribute modified versions.

ii) Open Source Software

Open Source Software is one in which, the *source code is made publicly available* to anyone. Users are typically granted the right to *view, modify, and distribute* the code under the terms of an open source license. Some examples are Ubuntu, Fedora, Android, Mozilla Firefox, Chromium, LibreOffice, Apache HTTP Server, WordPress, etc.

1.3.5 Closed Source Software

Closed Source Software, also known as *Proprietary Software* is the software whose *source code is kept confidential* and is only accessible to the original developers or the organization that owns it. Some examples are Microsoft Windows, Apple Mac-OS, Apple iOS, Word, Excel, PowerPoint, Adobe Creative Suite(Photoshop, Illustrator), Zoom, etc. Table 1.7 shows the features of Open -Source and Closed-software.

Table 1.7 - Features of open and closed source software

Feature	Open-Source Software	Closed-Source (Proprietary) Software
Source Code	Open. You can view, study, and modify it	Closed. Kept secret by the company
Updates/Fixes	Collaboration by a global community; fixes can be rapid	Only the developing company can apply fixes
Security	Transparent."Many eyes" principle, vulnerabilities are often found and fixed quickly	Obscure. Must trust the vendor that no backdoors or bugs exist
Customization	High. You can change the program to suit your needs	Low. You must wait for the company to add a feature

Richard Stallman - The Father of Free Software

Richard Matthew Stallman (born March 16, 1953), often known by his initials RMS, is an American computer programmer and the founder of the Free Software Movement. He is one of the most influential figures in the history of computing, advocating for the ethical right of users to control their own technology.

Early Life and Career

Stallman's journey began in the early 1970s at the MIT Artificial Intelligence Laboratory. During this time, programmers lived in a "hacker culture" where they freely shared and improved each other's code. However, by the early 1980s, software began to be 'privatised', meaning companies restricted access to the source code. Stallman found this ethically wrong, believing that software should be a community resource.

Major Contributions

Stallman is best known for three landmark contributions to the world of Information Technology.

The GNU Project (1983): *He launched this project to create a complete, Unix-like operating system that was entirely free. The name GNU is a recursive acronym for 'GNU is Not Unix.'*

The Free Software Foundation (FSF) (1985): *A non-profit organization established to support the GNU Project and promote the legal and philosophical principles of software freedom.*

GNU General Public License (GPL): *Stallman introduced the concept of Copyleft. Unlike traditional copyright that restricts users, the GPL uses law to ensure that the software remains free for everyone to use, modify, and redistribute.*

Try Yourself

1. _____ software manages the hardware and provides a platform for application software.
2. _____ is an example for open-source software.
3. Programs designed for specific tasks like word processing are called _____
4. FSF was founded by ----------.
5. Expand FOSS.
6. Distinguish between system software and application software.
7. Write the functions of an operating system.
8. Give two examples each for open source and free software.
9. Explain different freedoms related to software.
10. Explain the use of Device drivers.

Try Yourself

11. What do you mean by utility program? Give any two examples.
12. Explain the types of software with examples.
13. Describe the role of utility software in a computer system.
14. Differentiate between open source and free software with examples.

Match the following

1. Operating System	(a) Mozilla Firefox
2. Utility Software	(b) Spreadsheet
3. Application Software	(c) Windows
4. Open Source	(d) Disk Cleanup

Activity

Install any one open source software (including device drivers) in a computer.

1.4 Data and Information

Data and information are extremely important in the modern world of Artificial Intelligence (AI). They serve as the fundamental basis for knowledge, decision-making, and innovation across all sectors, including business, science, government, and even in daily life. In our daily activities, we encounter large quantities of raw facts—such as numbers, words, figures, or symbols. When these raw facts are scattered or unorganized, they lack specific meaning or context. However, they become useful once they are organized, structured, and processed. Computers are widely used for data processing because of their inherent speed, accuracy, and reliability. A computer system treats everything as data such as characters, pictures, audio, video, documents, and instructions. Most people use the terms data and information interchangeably, even though they are actually quite different. You will understand the critical distinction between data and information, the various stages of data processing, and the essential role computers play in this transformation. The following are the different stages of data transformation depicted in the pyramid diagram.



Pyramid diagram of data transformation

Data

Data is a raw, unorganized fact or a set of facts, such as numbers, text, images, audios, videos, etc. Data on its own has no complete meaning. For example, exam scores of a student, like 85, 92, 78, 95, are data.

Information

Processed data is called *information*. From the previous example, we can generate information such as total score or average score by processing the scores.

Knowledge

Knowledge is the ability to use information to make decisions. For example, by comparing a student's marks with the class average, we can understand whether the student needs improvement.

Intelligence

Intelligence is the ability to apply knowledge to solve problems in different situations. To improve the scores of students, apply strategies such as grouping them based on their learning styles and giving them different types of homework.

Wisdom

Wisdom is the ability to use knowledge and experience to make good decisions in real life situations. For example, a student notices that their maths score is low in two exams. Instead of feeling upset, he decide to change the study plan by practice daily, ask the teachers help, and revise weak topics.

1.4.1 Data Processing

Data processing is a series of activities carried out to convert raw data into meaningful information. It includes collection, manipulation and transformation of data into information. It involves a systematic set of operations performed on data to achieve a desired outcome. This process can be done manually or electronically.

i) Manual Data Processing (MDP)

It is performed using human effort with simple tools like a pen and paper. This method is slow and prone to errors.

ii) Electronic Data Processing (EDP)

It is the most common method today, which uses electronic devices like computers. It is highly efficient, fast, and accurate.

1.4.2 Stages of Data Processing

The data processing involves different stages that transform raw data into meaningful information. These stages are very important in decision-making. Modern data processing has evolved from simple '*collect-input-process-storage-output*' cycles into a complex, continuous process. Nowadays, the concept is defined by real-time intelligence, cloud architecture, and AI-driven automation. Data processing has evolved from a rigid, manual, and localised activity into an automated, global, and real-time operation. Instead of waiting for reports at the end of the day, modern systems process data as it happens, allowing the environment to respond to events in milliseconds. Stages of data processing are shown in Figure 1.9.

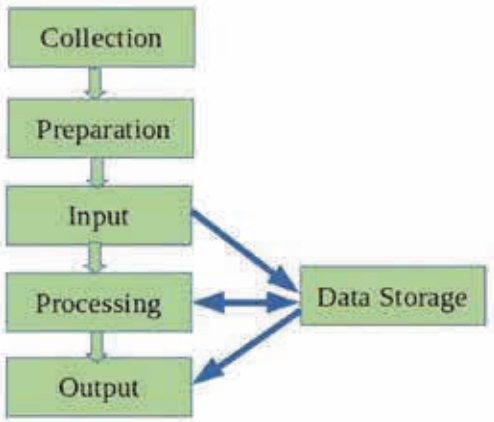


Figure 1.9. Stages of data processing

i. Collection

This is the first and most critical stage. Gathering raw data from various sources is called data collection. Data is collected from sensors, surveys, transaction logs or physical documents. This stage ensures that the data gathered is accurate and complete. For example, collecting marks of students, recording daily sales or registering patients in a hospital.

ii. Preparation

Raw data often contains errors or unnecessary information. In this stage, data is '*cleaned and organized*'. The main tasks of this stage are removing duplicates, correcting errors, and filtering out irrelevant facts (often called "Garbage In, Garbage Out" or GIGO). Here we ensure that only high-quality data moves to the next stage.

iii. Input

It is the activity of entering the collected data into the computer system using input devices and other strategies. The cleansed data is converted into a machine-readable format and entered into the computer system. For example, entering sales figures using a keyboard, barcode scanner or by importing digital files.

iv. Processing

This is the stage where the computer actually works on the data. The CPU performs various operations based on instructions.

Manipulating the input data according to given instructions is known as processing. Processing may involve calculations, comparisons, sorting, classifications, or summarising. For example, calculating total sales or generating grades from exam scores.

v. Storage

Storage is the mechanism that holds the data for each transformation phase in data processing. The processed data and results are stored here for future use. Data can be stored temporarily in primary memory (RAM) or permanently in secondary storage (hard disks, SSDs). Nowadays the storage types like cloud, database, archives, etc. are also in use.

vi. Output

Presenting the information in a human-readable form using output devices is called the output. This is the real stage to present the "Information" in a way that users can easily interpret for decision making. For example, displaying the reports, graphs, charts, tables on the monitor, printing an invoice or presenting audio/video.

1.4.3 Computer as a Data Processing Machine



Computer - a data processing machine

The different functional units of computer are used to perform the activities involved in data processing. Inputting can be done with keyboard/mouse, to store the data use devices like HDD, CD, Pen drive, etc., use CPU for processing and display output using monitor, projector, etc. Today computers are widely used for data processing because of their tremendous features. Some of which are the following.

Speed : Computers can process millions of instructions per second.

Accuracy : Errors are minimal if correct instructions and data are given.

Storage : Computers can store large volumes of data.

Diligence : The ability to work for hours without getting tired or bored.

Automation : Once a program is given, computers can perform tasks automatically.

Versatility : The ability to perform different kinds of work. It is used in every walk of human life.



1. Raw facts and figures are called _____.
2. Processed data is called ----------
3. _____ stage converts data into meaningful information.
4. Define data and information. Give examples.
5. List the different stages of data processing.
6. Computers are called data processing machines. Why ?
7. Differentiate between data and information.
8. Explain the stages of data processing.
9. Briefly explain the features of a computer system.
10. What are the different types of data processing?
11. Explain manual and electronic data processing.

Match the following

1. Data	(a) Organised and meaningful
2. Information	(b) Raw facts and figures
3. Input	(c) Keeping processed results
4. Storage	(d) Entering data

Activity/Case study

List and write the steps involved in plus one admission procedure (SWS) through hscap.

1.5 Green Computing

Computers and electronic devices have become an integral part of our daily life. However, their tremendous use has created two major concerns-environmental impact of computing and electronic waste (e-Waste) management. By the end of this session, you will learn about green computing and the problems posed by e-waste, as well as the various methods for its disposal.

To reduce harmful effects on the environment, the concepts of Green Computing and proper e-Waste disposal have become very important. Green computing, also known as green IT, is the practice of designing, manufacturing, using, and disposing of computers and related subsystems (like monitors, printers, and



Green Computing

storage devices) in an environment (eco) friendly and energy-efficient way. The goal is to minimize the negative impact of technology on the environment. Many corporate IT departments have green computing initiatives to reduce the environmental effect of their IT operations. Computers and electronic devices are to be used responsibly to support sustainable development. Through Green Computing and safe disposal of e-Waste, we can ensure that technology benefits society without harming the environment.

The key principles of green computing include

Energy Efficiency: Energy efficiency is the practice of using less power for devices and data centres. This involves using energy-saving hardware and implementing power management features.

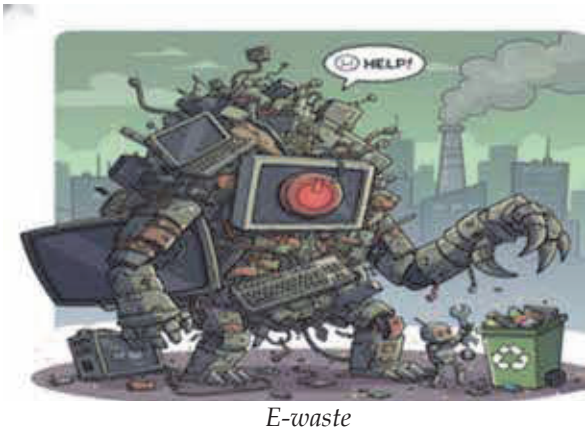
Eco-Friendly Materials: Designing and manufacturing devices with biodegradable materials that are less harmful to the environment and which can be recycled.

Waste Reduction: Extending the lifespan of devices to reduce the amount of electronic waste produced. This is a key focus area.

Responsible Disposal: Ensuring that old and obsolete equipment is handled and disposed properly and safely.

1.5.1 E-waste

E-waste, or electronic waste, refers to discarded electrical or electronic devices such as computers, mobile phones, televisions, printers, and other gadgets. As technology rapidly evolves, consumers and manufactures frequently upgrade devices like smart phones, laptops, and televisions, creating a massive amount of e-waste.



E-waste

The main problem with e-waste is its composition. It contains both valuable materials like gold, silver, copper, and palladium, and hazardous substances such as lead, mercury, cadmium, bromine and arsenic. When e-waste is improperly disposed of, the toxic substances can dissolve into the soil and groundwater, contaminating ecosystems and posing serious health risks to humans and animals.

1.5.2 E-waste Disposal Methods

Responsible disposal of e-waste is crucial for protecting the environment and human health. Here are the main e-waste disposal methods, including both safe and harmful or unsafe.

i) Safe Methods

Reuse: If a device is still functioning after its long term usage, sell it, or give it to someone (Schools, Individuals in need, etc.) who can use it.

Recycle: For devices that are no longer working, ensure they are sent to a certified e-waste recycling facility. These facilities can safely dismantle devices, separate hazardous materials, and recover valuable metals and plastics for use in new products.

Manufacturer Take-Back Programmes: Many electronics companies (e.g., Apple, Dell and HP) offer free return trade-in programme.



E-Waste disposal

ii) Unsafe Methods

Land filling: E-waste is simply buried in landfills, where its toxic chemicals can contaminate the surrounding environment.

Incineration: Burning e-waste at high temperatures reduces its volume but can release harmful toxins and gases into the atmosphere.

Acid Baths: In some informal recycling operations, components are submerged in strong acids to dissolve metals. This is highly dangerous and produces toxic chemical waste that is often improperly disposed of, polluting local water sources.

Exporting to Developing Countries: Illegal or unethical export of e-waste to countries with poor regulations leads to environmental pollution and health hazards for workers handling e-waste without protection.

1.5.3 What Should Be Done

To protect ecosystems from e-waste damage, we need to

Recycle electronic devices responsibly via certified recyclers.

Reduce e-waste by buying less or repairing devices.

Proper disposal of e-waste with drop-off centres or manufacturer take-back.

Educate people to handle the e-waste safely.



1. The eco-friendly use of computers is called _____.
2. Discarded electronic devices are known as _____.
3. Toxic substances like mercury and lead are found in _____.
4. Define Green Computing. Write about any two green computing practices .
5. What are the health hazards caused by e-Waste?
6. Distinguish between reuse and recycling.
7. What are the key principles of green computing?
8. Why do we protect our ecosystems ? Explain.
9. Explain the concept of e-Waste and its problems.
10. Discuss the disposal methods of e-Waste with examples.
11. How does Green Computing contribute to environmental protection?

Match the following

1. Green Computing	(a) Discarded computers
2. e-Waste	(b) Energy-efficient use of technology
3. Recycling	(c) Producer responsibility
4. Take-back programmes	(d) Extracting valuable materials from old devices

Activity

Collect the e-wastes in your school and segregate them into reusable, recyclable, and unusable categories and suggest appropriate e-waste disposal methods for each of them.

Summary

A computing system is a programmable electronic machine that transforms raw data into information. Most modern systems follow the Von Neumann architecture, which utilises a Central Processing Unit (CPU) to fetch and execute instructions stored alongside data in a single memory unit. Data processing has evolved significantly from early manual tools like the Abacus to high-speed electronic machines which are capable to handle complex tasks in fractions of a second. Beyond hardware, the system depends on software, which is broadly classified into system software

and application software. Free and Open-Source Software (FOSS) is a vital part of this ecosystem, granting users the freedom to run, study, modify, and redistribute the source code. In contrast, closed-source or proprietary software keeps its source code confidential and freedoms are restricted to the users. The rapid pace of development means that new technologies emerge quickly which often leads to significant environmental challenges such as the accumulation of e-waste. E-waste consists of discarded devices like old computers and smart phones that can release toxic substances if not handled correctly. To tackle this situation effectively, green computing practices emphasis on energy efficiency and the responsible recycling or reuse of electronic components to protect ecosystems. As a whole this chapter covers the technological developments from historical roots to the modern organization of computing systems.



1. ROM stands for _____ .
2. Data processing converts raw data into _____.
3. Removing duplicate records is done during _____ stage.
4. Which unit acts as a 'traffic policeman' by directing the flow of data between functional units?
   a) Arithmetic Logic Unit b) Accumulator c) Control Unit d) Status Register
5. In the memory hierarchy, which unit is exactly equal to half a byte?
   a) Bit b) Nibble c) Kilo byte d) Peta byte
6. Which of the following is considered a "hardcopy" output device?
   a) Monitor b) Projector c) Printer d) Speaker
7. Which organization was founded by Richard Stallman in 1985 to promote software ethics?
   a) OSI b) FSF c) KITE d) GNU Project
8. Which stage of data processing removes errors and duplicates?
   a) Collection b) Input c) Preparation d) Output
9. Which unit of the computer performs processing?
   a) Input unit b) Output unit c) Storage unit d) CPU
10. "Garbage In, Garbage Out (GIGO)" is related to which stage?
    a) Collection b) Preparation c) Storage d) Output
11. Which of the following is a hazardous substance found in e-waste?
    a) Oxygen b) Mercury c) Nitrogen d) Helium
12. Which is a safe method of e-waste disposal?
    a) Incineration b) Land filling c) Recycling d) Acid bath

13. Manufacturer take-back programmes are offered by
    a) Schools b) Hospitals c) Electronics companies d) Farmers
14. Distinguish between General-purpose and Special-purpose application software.
15. Explain the hierarchy and relationship between the CPU, Cache memory, and RAM.
16. How does Cache memory improve CPU utilization?
17. Compare Open source and Closed source softwre.
18. What is the primary function of a "Device Driver"?
19. Give any two examples of "Ambient gadgets ".
20. What are the three levels of Cache memory?
21. List the different functions of an Operating System.
22. Explain the modern Input strategy and the "Always-Available" Concept.
23. Explain the features of computer system.
24. Define data processing.
25. Write the advantage of electronic data processing.
26. Write any two differences between MDP and EDP.
27. List the stages of data processing.
28. Write any two advantages of computers in data processing.
29. What is meant by storage in data processing?
30. Give two examples of data collection.
31. Explain Manual Data Processing and Electronic Data Processing with examples.
32. Explain any four features of computers as a data processing machine
33. Explain the role of a computer as a data processing machine.
34. Draw and explain the Stages of Data Processing.
35. Define e-waste.
36. Name any two hazardous substances found in e-waste.
37. Give two examples of an electronic device that becomes e-waste.
38. Write any one objective of green computing.
39. What is a manufacturer take-back programme?
40. List any two principles of green computing.
41. Write any two problems caused by improper e-waste disposal.
42. Write any two safe methods of e-waste disposal.
43. What are the hazardous substances present in e-waste?
44. Explain green computing in detail. Discuss its key principles.
45. Describe the safe and unsafe methods of e-waste disposal.